

Surface Mount Type SP-Cap

Series : **GX (Super Low ESR Products)**



Features

- Large capacitance (560 μF max.)
- Super Low ESR (3 $\text{m}\Omega$ max.)
- Low ESL (3-terminals : 50 % less than 2-terminals) [Suffix : L]
- High ripple current (10200 $\text{mA}_{\text{R.M.S.}}$ max.)
- RoHS compliance, Halogen free

Specifications

Series	GX	
Category temp. range	-55 °C to +105 °C	
Rated voltage range	2 V.DC to 2.5 V.DC	
Nominal cap.range	330 μF to 560 μF	
Capacitance tolerance	$\pm 20\%$ (120 Hz/+20 °C)	
DC leakage current	$I \leq 0.1 \text{ CV}$ (μA) 2 minutes	
Dissipation factor ($\tan \delta$)	≤ 0.06 (120 Hz/+20 °C)	
Surge voltage (V.DC)	Rated voltage $\times 1.25$ (15 °C to 35 °C)	
Endurance	+105 °C, 2000 h, rated voltage applied	
	Capacitance change	Within $\pm 20\%$ of the initial value
	$\tan \delta$	≤ 2 times of the initial limit
	DC leakage current	≤ 3 times of the initial limit
Damp heat (Steady state)	+60 °C, 90 %, 500 h, No-applied voltage	
	Capacitance change of initial measurd value	2 V.DC to 2.5 V.DC +70 %, -20 %
	$\tan \delta$	≤ 2 times of the initial limit
	DC leakage current	Within the initial limit

Marking

		Rated voltage mark <table><tr><td>d</td><td>2 V.DC</td></tr><tr><td>e</td><td>2.5 V.DC</td></tr></table>		d	2 V.DC	e	2.5 V.DC
d	2 V.DC						
e	2.5 V.DC						

Dimensions (not to scale)

2 terminals

Unit : mm

Series	L ± 0.2	W1 ± 0.2	W2 ± 0.1	H ± 0.1	P ± 0.3
GX	7.3	4.3	2.4	1.9	1.3

* Externals of figure are the reference.

3 terminals

Unit : mm

Series	L ± 0.2	W1 ± 0.2	W2 ± 0.1	H ± 0.1	P1 ± 0.3	P2 ± 0.1	P3 ± 0.2	P4 ± 0.2
GX	7.3	4.3	2.4	1.9	1.3	1.1	0.7	1.4

* Externals of figure are the reference.

Characteristics list

Reflow *3										<Standard>	
Series	Rated voltage (V.DC)	Capacitance (±20 %) (μF)	Case size (mm)			Specification		The number of terminals		Part number	Min.*4 Packaging Q'ty (pcs)
			L	W	H	*1 Ripple current (mAr.m.s.)	*2 ESR (mΩ max.)	2	3		
GX	2	330	7.3	4.3	1.9	10200	3	○		EEFGX0D331R	3500
		470	7.3	4.3	1.9	10200	3	○		EEFGX0D471R	3500
			7.3	4.3	1.9	10200	3		○	EEFGX0D471L	3500
		560	7.3	4.3	1.9	10200	3	○		EEFGX0D561R	3500
			7.3	4.3	1.9	10200	3		○	EEFGX0D561L	3500
	NEW	330	7.3	4.3	1.9	10200	3	○		EEFGX0E331R	3500
	2.5	470	7.3	4.3	1.9	10200	3	○		EEFGX0E471R	3500
			7.3	4.3	1.9	10200	3		○	EEFGX0E471L	3500

*1: Ripple current (100 kHz/ +45°C), *2: ESR (100 kHz/+20 °C)

*3: Please refer to the page of "Mounting Specifications".

*4: Please contact us when 500 pcs packing is necessary.

Temperature compensation multipliers for ripple current

Temp.	T ≤ 45 °C	45 °C < T ≤ 85 °C	85 °C < T ≤ 105 °C
Coefficient	1.0	0.7	0.25

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